

Question Paper: Ai

****Artificial Intelligence Question Paper for Class 9-12****

****Medium Difficulty****

****Multiple Choice Questions (1-10)****

1. What is the primary goal of Artificial Intelligence?

- A) To develop machines that can think like humans
- B) To develop machines that can only perform repetitive tasks
- C) To develop machines that can only understand human language
- D) To develop machines that can only recognize images

2. Which of the following is an application of AI in healthcare?

- A) Virtual assistants
- B) Chatbots
- C) Medical diagnosis and treatment
- D) Language translation

3. What is the term for the process of teaching a machine to make predictions based on data?

- A) Machine learning
- B) Deep learning
- C) Natural language processing
- D) Computer vision

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4. Which of the following AI techniques is used for image recognition?

- A) Support Vector Machines
- B) Convolutional Neural Networks
- C) Recurrent Neural Networks
- D) Decision Trees

5. What is the difference between narrow and general AI?

- A) Narrow AI is more intelligent than general AI
- B) General AI is more intelligent than narrow AI
- C) Narrow AI is designed to perform a specific task, while general AI can perform any task
- D) Narrow AI can only understand human language, while general AI can understand any language

6. Which of the following is a type of machine learning algorithm?

- A) Linear regression
- B) Clustering
- C) Decision trees
- D) All of the above

7. What is the purpose of the backpropagation algorithm in neural networks?

- A) To update the weights and biases of the network
- B) To initialize the weights and biases of the network
- C) To normalize the input data
- D) To regularize the output data

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8. Which of the following is an example of a natural language processing task?

- A) Sentiment analysis
- B) Text classification
- C) Language translation
- D) All of the above

9. What is the term for the field of study that focuses on developing machines that can interact with humans in a more human-like way?

- A) Human-computer interaction
- B) Artificial intelligence
- C) Robotics
- D) Cognitive science

10. Which of the following is a challenge in developing AI systems?

- A) Data quality
- B) Model complexity
- C) Interpretability
- D) All of the above

****Short Questions (11-15)****

11. What is machine learning, and how does it differ from traditional programming? (3-5 marks)

12. Describe the concept of neural networks and their application in AI. (3-5 marks)

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13. What is the difference between supervised and unsupervised learning? Provide an example of each. (3-5 marks)

14. Explain the concept of deep learning and its advantages over traditional machine learning. (3-5 marks)

15. Describe the application of AI in robotics and its potential impact on society. (3-5 marks)

****Long/Descriptive Questions (16-18)****

16. Discuss the concept of artificial general intelligence (AGI) and its potential implications on society. Describe the current state of AGI research and the challenges that need to be addressed. (8-10 marks)

17. Explain the concept of natural language processing (NLP) and its application in AI. Describe the different NLP tasks, such as sentiment analysis, text classification, and language translation. Provide examples of each task and discuss the challenges associated with NLP. (8-10 marks)

18. Describe the concept of computer vision and its application in AI. Explain the different computer vision tasks, such as image recognition, object detection, and image segmentation. Provide examples of each task and discuss the challenges associated with computer vision. (8-10 marks)

****Answer Key****

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****Multiple Choice Questions****

1. A) To develop machines that can think like humans
2. C) Medical diagnosis and treatment
3. A) Machine learning
4. B) Convolutional Neural Networks
5. C) Narrow AI is designed to perform a specific task, while general AI can perform any task
6. D) All of the above
7. A) To update the weights and biases of the network
8. D) All of the above
9. A) Human-computer interaction
10. D) All of the above

****Short Questions****

11. Machine learning is a type of AI that involves training machines to learn from data, whereas traditional programming involves writing rules and instructions for the machine to follow.
12. Neural networks are a type of machine learning model inspired by the structure and function of the human brain. They are applied in AI for tasks such as image recognition, natural language processing, and decision-making.
13. Supervised learning involves training a machine on labeled data, whereas unsupervised learning

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involves training a machine on unlabeled data. Examples of supervised learning include image classification, while examples of unsupervised learning include clustering.

14. Deep learning is a type of machine learning that involves the use of multiple layers of neural networks. Its advantages include the ability to learn complex patterns in data and improve performance over time.

15. AI is applied in robotics for tasks such as navigation, object recognition, and human-robot interaction. Its potential impact on society includes the development of autonomous vehicles, smart homes, and healthcare robots.

****Long/Descriptive Questions****

16. AGI refers to the development of machines that can perform any intellectual task that a human can. The current state of AGI research is focused on developing machines that can learn, reason, and interact with humans in a more human-like way. Challenges that need to be addressed include the development of more advanced machine learning algorithms and the integration of multiple AI systems.

17. NLP is a field of study that focuses on the interaction between computers and humans in natural language. NLP tasks include sentiment analysis, text classification, and language translation. Challenges associated with NLP include the ambiguity of human language, the need for large amounts of training data, and the development of more advanced machine learning algorithms.

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18. Computer vision is a field of study that focuses on the development of machines that can interpret and understand visual data from the world. Computer vision tasks include image recognition, object detection, and image segmentation. Challenges associated with computer vision include the complexity of visual data, the need for large amounts of training data, and the development of more advanced machine learning algorithms.